

OASIS INFOBYTE

DATA SCIENCE INTERNSHIP

TASK 1 – IRIS FLOWER CLASSIFICATION

Project Information	Details
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Internship Program	Oasis Infobyte SIP
Track	Data Science
Task	Task 1 – Iris Flower Classification
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Primary Platform	Jupyter Notebook
Dataset	Scikit-learn Iris Dataset

Table of Contents

- 1. Executive Summary**
- 2. Project Objective**
- 3. Problem Statement**
- 4. Dataset Description**
- 5. Tools, Technologies and Libraries**
- 6. Project Workflow**
- 7. Dataset Loading and Initial Inspection**
- 8. DataFrame Creation and Target Encoding**
- 9. Exploratory Data Analysis**
- 10. Descriptive Statistics**
- 11. Species Distribution Analysis**
- 12. Pairplot Analysis**
- 13. Box Plot Analysis**
- 14. Correlation Analysis and Feature Selection**
- 15. Train-Test Split**
- 16. Feature Scaling**
- 17. Logistic Regression Model**
- 18. Logistic Regression Evaluation**
- 19. K-Nearest Neighbors Model**
- 20. KNN Evaluation**
- 21. Model Comparison**
- 22. Final Prediction**
- 23. Conclusion**
- 24. Limitations and Possible Improvements**

1. Executive Summary

This report documents the completion of Task 1 of the Oasis Infobyte Data Science Internship. The task focuses on Iris flower classification using supervised machine learning. The project uses the built-in Iris dataset available through Scikit-learn and follows a complete data science workflow: dataset loading, data preparation, exploratory data analysis, visualization, feature analysis, train-test splitting, feature scaling, model training, model evaluation, model comparison, and prediction.

Two classification algorithms were implemented: Logistic Regression and K-Nearest Neighbors (KNN). The dataset was divided into 80% training data and 20% testing data, producing 120 training records and 30 testing records. StandardScaler was applied to the numerical features before model training.

Both models achieved an accuracy of 93.33% on the test dataset. Since the two models achieved the same accuracy, the project does not identify one as superior using accuracy alone. The trained Logistic Regression model was then used to classify a new Iris flower and predicted the species as Versicolor.

2. Project Objective

The objective of the task is to build a machine learning classification model capable of identifying the species of an Iris flower based on its physical measurements. The project also demonstrates the practical steps required to prepare data, investigate patterns, train classification algorithms, evaluate results, and communicate findings.

- Load and inspect the Iris dataset.
- Convert the dataset into a structured Pandas DataFrame.
- Convert numerical target labels into readable species names.
- Check dimensions, data types, missing values, and descriptive statistics.
- Perform exploratory data analysis using charts and statistical relationships.
- Identify features that provide useful separation between species.
- Split the data into training and testing sets using an 80/20 ratio.
- Standardize the numerical features using StandardScaler.
- Train Logistic Regression and KNN classifiers.
- Evaluate the models using accuracy, precision, recall, F1-score, and confusion matrices.
- Compare model performance.
- Use the trained Logistic Regression model to make a new prediction.

3. Problem Statement

Given four measurements of an Iris flower—sepal length, sepal width, petal length, and petal width—the machine learning system must predict which of three species the flower belongs to: Setosa, Versicolor, or Virginica. This is a multi-class supervised classification problem because the target variable represents three discrete categories.

4. Dataset Description

The project uses the Iris dataset provided by Scikit-learn. The dataset contains measurements from three Iris species. Each observation has four numerical input features and one target class.

Feature / Target	Type	Description
Sepal Length (cm)	Numerical	Length of the sepal in centimeters.

Sepal Width (cm)	Numerical	Width of the sepal in centimeters.
Petal Length (cm)	Numerical	Length of the petal in centimeters.
Petal Width (cm)	Numerical	Width of the petal in centimeters.
Species	Categorical	Setosa, Versicolor, or Virginica.

The dataset contains 150 observations and four numerical input features. Each species contributes 50 observations, so the class distribution is balanced.

Dataset Property	Value
Total observations	150
Input features	4
Target classes	3
Observations per class	50
Training observations	120
Testing observations	30

5. Tools, Technologies and Libraries

Technology / Library	Purpose in the Project
Python	Main programming language used for the analysis and machine learning workflow.
Pandas	DataFrame creation, inspection, descriptive analysis, and data manipulation.
NumPy	Numerical array operations and preparation of new prediction input.
Matplotlib	Creation and display of visualizations.
Seaborn	Statistical plots including count plots, pairplots, box plots, and heatmaps.
Scikit-learn	Dataset loading, train-test split, scaling, classification models, and evaluation metrics.
Jupyter Notebook	Interactive development environment used to document code, outputs, observations, and results.

6. Project Workflow

The implementation followed the sequence below:

1. Import required libraries.
2. Load the built-in Iris dataset.
3. Inspect feature names, target names, and dataset dimensions.

4. Convert the dataset into a Pandas DataFrame.
5. Convert numerical target labels into species names.
6. Perform EDA: shape, data types, missing values, and descriptive statistics.
7. Analyze species distribution.
8. Create and interpret a pairplot.
9. Create and interpret box plots.
10. Calculate and visualize the feature correlation matrix.
11. Prepare X and y variables.
12. Split data into 80% training and 20% testing sets.
13. Scale the features with StandardScaler.
14. Train Logistic Regression.
15. Evaluate Logistic Regression.
16. Train KNN with five neighbors.
17. Evaluate KNN.
18. Compare model accuracies.

7. Dataset Loading and Initial Inspection

The Iris dataset was loaded using Scikit-learn's `load_iris()` function. The notebook confirmed that the dataset contains four features—sepal length, sepal width, petal length, and petal width—and three target names: setosa, versicolor, and virginica. The original dataset shape was (150, 4), representing 150 observations across four features.

```
from sklearn.datasets import load_iris

iris = load_iris()
print(iris.feature_names)
print(iris.target_names)
print(iris.data.shape)
```

8. DataFrame Creation and Target Encoding

The numerical feature matrix was converted into a Pandas DataFrame. A new 'species' column was added using the original numerical target values. The numerical target values were then mapped to readable labels: 0 to setosa, 1 to versicolor, and 2 to virginica. This made the dataset easier to interpret during analysis and visualization.

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df["species"] = iris.target

df["species"] = df["species"].map({
    0: "setosa",
    1: "versicolor",
    2: "virginica"
})
```

9. Exploratory Data Analysis

Exploratory Data Analysis was used to understand the structure and characteristics of the dataset before model training. The analysis checked the number of rows and columns, data types, missing values, descriptive statistics, class distribution, relationships among numerical variables, and differences in feature distributions across species.

10. Descriptive Statistics

The descriptive statistics summarized the numerical features using count, mean, standard deviation, minimum, first quartile, median, third quartile, and maximum values.

Feature	Mean	Std. Dev.	Min	Median	Max
Sepal Length	5.8433	0.8281	4.3	5.8	7.9
Sepal Width	3.0573	0.4359	2.0	3.0	4.4
Petal Length	3.7580	1.7653	1.0	4.35	6.9
Petal Width	1.1993	0.7622	0.1	1.3	2.5

The statistics show that petal length and petal width have greater variability than the sepal measurements. The wide ranges of petal measurements also support the later visual observation that petal features provide stronger separation among species.

11. Species Distribution Analysis

A count plot was created to examine the number of records belonging to each species. The output showed 50 Setosa, 50 Versicolor, and 50 Virginica observations. Therefore, there is no class imbalance in the dataset.

A balanced target distribution is useful because each class contributes equally to the model evaluation. It reduces the risk that a model's overall accuracy is dominated by a much larger class.

12. Pairplot Analysis

A Seaborn pairplot was used to visualize pairwise relationships among the four numerical features, with species used as the hue. The purpose was to see whether observations belonging to different species form distinct groups.

The pairplot showed clearer separation for petal length and petal width. The Setosa observations were particularly distinct from the other two species in petal measurements. Versicolor and Virginica showed more overlap, although their petal measurements still provided useful separation. The sepal measurements showed comparatively more overlap.

13. Box Plot Analysis

Box plots were created for each numerical feature with species on the x-axis. Box plots make it possible to compare the median, spread, and potential outlying observations for each feature across the three species.

Petal length and petal width showed the strongest differences among species, with relatively little overlap. Sepal length also showed noticeable differences, whereas sepal width had more overlap. Based on this

analysis, petal length and petal width were identified as especially useful measurements for distinguishing species.

14. Correlation Analysis and Feature Selection

A correlation matrix was calculated for the four numerical features. Correlation values range from -1 to +1, with values closer to +1 indicating a strong positive linear relationship and values closer to -1 indicating a strong negative relationship.

Feature Pair	Correlation
Sepal Length – Sepal Width	-0.1176
Sepal Length – Petal Length	0.8718
Sepal Length – Petal Width	0.8179
Sepal Width – Petal Length	-0.4284
Sepal Width – Petal Width	-0.3661
Petal Length – Petal Width	0.9629

The strongest relationship was between petal length and petal width, with a correlation of approximately 0.963. This indicates that these two measurements tend to increase together. Importantly, feature selection in this project was informed by the combination of correlation analysis, pairplot separation, and box-plot distributions rather than correlation alone.

15. Train-Test Split

Before training the models, the input features were separated from the target variable. The features were stored in X and the species labels in y. The data was then split using `train_test_split` with `test_size=0.20`, `random_state=42`, and `stratify=y`.

Subset	Number of Rows	Percentage
Training set	120	80%
Testing set	30	20%

The `stratify=y` parameter preserves the class proportions during the split. The fixed `random_state=42` makes the split reproducible when the notebook is run again with the same data and code.

16. Feature Scaling

`StandardScaler` was applied to the numerical features. Scaling transforms the features so that they are on a comparable standardized scale. This is particularly useful for algorithms such as KNN because distance calculations can otherwise be influenced by differences in feature scale.

```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

The scaler was fitted only on the training data and then used to transform both training and testing data. This prevents information from the test set from influencing the scaling parameters and helps avoid data leakage.

17. Logistic Regression Model

Logistic Regression was selected as the first classification model. It is a supervised classification algorithm that can predict categorical classes from numerical input features. In this project, the model was trained on the standardized four-feature input. The `max_iter` parameter was set to 200 to provide sufficient iterations for the model optimization process.

18. Logistic Regression Evaluation

The Logistic Regression model was evaluated using accuracy, classification report, and confusion matrix.

Class	Precision	Recall	F1-score	Support
Setosa	1.00	1.00	1.00	10
Versicolor	0.90	0.90	0.90	10
Virginica	0.90	0.90	0.90	10
Overall Accuracy	—	—	0.93	30

The model achieved 93.33% accuracy, meaning 28 of the 30 test observations were correctly classified. Setosa was classified perfectly in the test set. Versicolor and Virginica each achieved 90% precision, recall, and F1-score.

The confusion matrix therefore indicates two total errors. The results demonstrate strong performance but also show that some overlap exists between Versicolor and Virginica.

19. K-Nearest Neighbors Model

K-Nearest Neighbors (KNN) was selected as the second classifier. KNN predicts the class of an observation based on the classes of nearby observations in the feature space. Because KNN relies on distances, the scaled features were used. The project used five nearest neighbors (`n_neighbors=5`). The same training and testing observations were used so that the KNN results could be compared fairly with Logistic Regression.

20. KNN Evaluation

The KNN model was evaluated using accuracy, classification report, and confusion matrix.

Class	Precision	Recall	F1-score	Support
Setosa	1.00	1.00	1.00	10
Versicolor	0.83	1.00	0.91	10
Virginica	1.00	0.80	0.89	10
Overall Accuracy	—	—	0.93	30

KNN achieved 93.33% accuracy, matching Logistic Regression. Setosa was classified perfectly. All Versicolor test observations were recalled, while two Virginica observations were classified as Versicolor. This explains the 1.00 recall for Versicolor, the 0.83 precision for Versicolor, and the 0.80 recall for Virginica.

21. Model Comparison

Model	Test Accuracy	Result
Logistic Regression	93.33%	28/30 correct
K-Nearest Neighbors	93.33%	28/30 correct

The accuracy comparison shows an exact tie. Both Logistic Regression and KNN achieved 0.933333 accuracy on the 30-sample test set. Therefore, the project should not claim that Logistic Regression is more accurate than KNN. Based on accuracy alone, both models have equal performance.

Although the two models have the same overall accuracy, their error patterns differ. Logistic Regression produced balanced performance across Versicolor and Virginica, while KNN had stronger Versicolor recall but lower Virginica recall. Consequently, if a future version of the project needs to select one model, additional criteria such as class-specific recall, F1-score, cross-validation, model simplicity, or computational requirements should be considered.

22. Final Prediction

The project used the trained Logistic Regression model to classify a new flower. The input measurements were:

Measurement	Value
Sepal Length	5.8 cm
Sepal Width	3.0 cm
Petal Length	4.5 cm
Petal Width	1.5 cm

Final prediction: VERSICOLOR

The model predicted that the new Iris flower belongs to the Versicolor species based on the supplied measurements.

23. Conclusion

This project successfully implemented an end-to-end supervised machine learning workflow for Iris flower classification. The work began with loading the Scikit-learn Iris dataset and preparing it for analysis. The target labels were converted into readable species names, and the dataset was inspected for structure, data types, missing values, and descriptive statistics.

EDA showed that the dataset is balanced across Setosa, Versicolor, and Virginica. The pairplot and box plots demonstrated that petal length and petal width provide the clearest separation between the species. The correlation matrix also showed a very strong positive relationship between petal length and petal width.

For machine learning, the dataset was split into 120 training records and 30 testing records using an 80/20 split. StandardScaler was fitted on the training data and applied to both datasets. Logistic Regression and KNN were then trained and evaluated using the same test set.

Both classifiers achieved 93.33% test accuracy. The equal accuracy means that neither model can be declared the best based on accuracy alone. The final Logistic Regression prediction for a new flower with measurements 5.8, 3.0, 4.5, and 1.5 was Versicolor.

24. Limitations and Possible Improvements

The project is successful for the internship task, but there are several ways the analysis could be strengthened in a future version:

- Evaluate additional classifiers such as Decision Tree and Random Forest, which were imported in the notebook but not included in the final model comparison.
- Use k-fold cross-validation instead of relying on a single train-test split.
- Compare additional metrics and class-specific performance before selecting a final model.
- Tune KNN's number of neighbors using validation rather than selecting five neighbors directly.
- Use a confusion-matrix analysis for each candidate model and examine error patterns.
- Create a consolidated model comparison table containing accuracy, precision, recall, and F1-score.
- Add reproducible environment information such as the Python and Scikit-learn versions used.

These improvements are optional extensions and were not necessary for the completed Task 1 workflow.